

Session 1 & 2: Machine Learning

Introduction, Categories, Machine Learning System Lifecycle (CO1)



Dr. G. Omprakash

KLEF Hyderabad (Aziz Nagar) – Dept. of ECE

Machine Learning (25SC2107E)

- 1 Introduction
- 2 Categories
- 3 The Lifecycle Stages

Introduction

What is Machine Learning?

Two classic definitions

Arthur Samuel (1959)

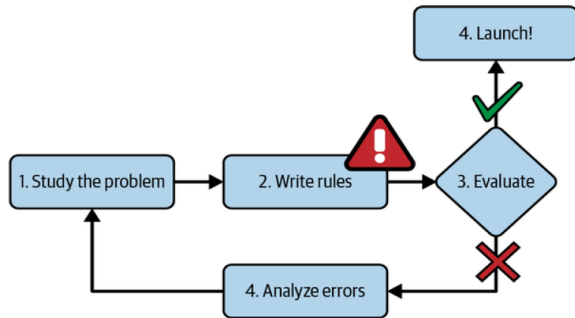
Machine learning is the field of study that gives computers the ability to learn without being explicitly programmed.

Tom Mitchell (1997)

A computer program is said to learn from experience **E** with respect to some task **T** and some performance measure **P**, if its performance on **T**, as measured by **P**, improves with experience **E**.

Machine learning is a subset of artificial intelligence where algorithms analyze data to learn patterns and make predictions or decisions without being explicitly programmed.

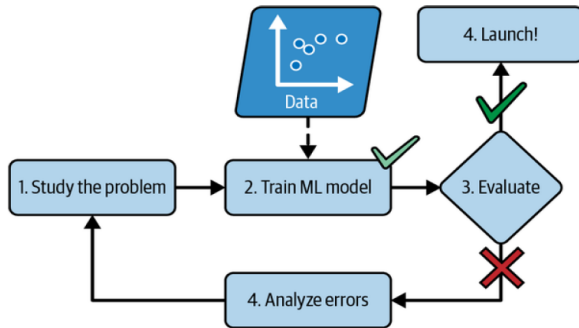
Traditional Approach (Spam filter)



Source: Geron

- First examine what spam typically looks like
 - Notice for phrases (such as “4U”, “credit card”, “free”, and “amazing”)
 - Notice a few other patterns: sender’s name, the email’s body, other parts of the email
- Write a detection algorithm for each of the patterns that you noticed
- Program would flag emails as spam if a number of these patterns were detected
 - Test your program and repeat steps 1 and 2 until it was good enough to launch.
- **Difficulty: Program will likely become a long list of complex rules**

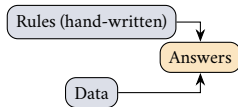
Machine Learning Approach (Spam filter)



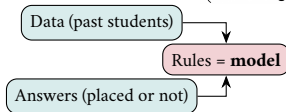
Source: Geron

- Spam filter based on machine learning techniques automatically learns which words and phrases are good predictors of spam
 - Detect unusually frequent patterns of words in the spam examples
- The program is much shorter, easier to maintain, and most likely more accurate

Traditional programming



Machine learning



Flip in paradigm

Traditional	Machine Learning
rules + data → answers	data + answers → rules
human writes logic	algorithm infers logic

Key Insight

ML inverts programming: instead of coding the decision logic, we let an algorithm *learn the logic from historical examples*.

AI vs ML vs Deep Learning

- **AI:** broad goal — machines that act intelligently.
- **ML:** a subset of AI — systems that learn from data.
- **Deep Learning:** a subset of ML — many-layered neural networks.
- This course is *classical* ML (linear models, trees, clustering); deep learning is a later course.

Categories

Taxonomy

Supervised
labelled examples (x, y)

Predict placement,
salary, spam, disease

Unsupervised
only inputs x

Group similar stu-
dents, compress features

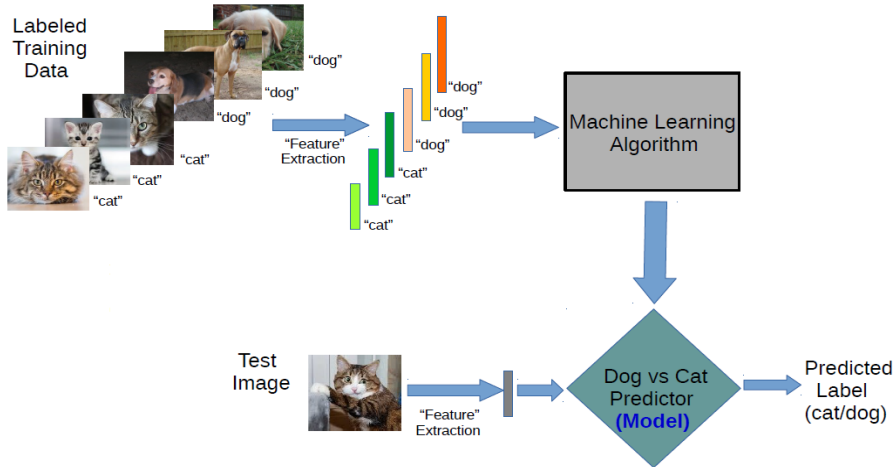
Reinforcement
actions $\rightarrow$ rewards

Game playing, robotics, ad bidding

- **Supervised:** learn a mapping $f : x \mapsto y$ from examples where the “right answer” y is known.
 - Eg: Classification, Regression, Ranking
- **Unsupervised:** no answers given — discover structure (clusters, low-dimensional patterns).
 - Eg: Clustering, Dimensionality Reduction
- **Reinforcement:** an agent learns by trial, error, and delayed reward.
 - RL doesn't use “labeled” or “unlabeled” data in the traditional sense!
 - An agent learns via its Interactions with an environment

Many other specialized flavors of ML also exist: Semi-supervised Learning, Active Learning, Transfer Learning, Multitask Learning

A Typical Supervised Learning Workflow (for classification)

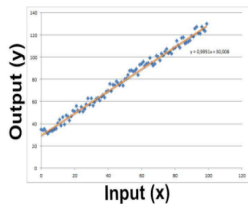


Source: CS771A (Piyush Rai)

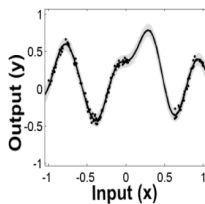
Supervised learning- Geometric View

Regression: Learn a line/curve (the “model”) using training data consisting of Input-output pairs.

- Each output is a real-valued number
- Use it to predict the outputs for new “test” inputs

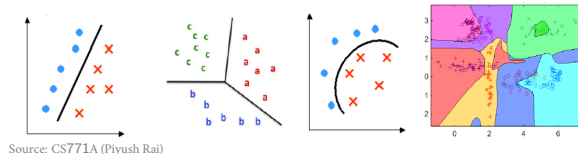


Source: CS771A (Piyush Rai)



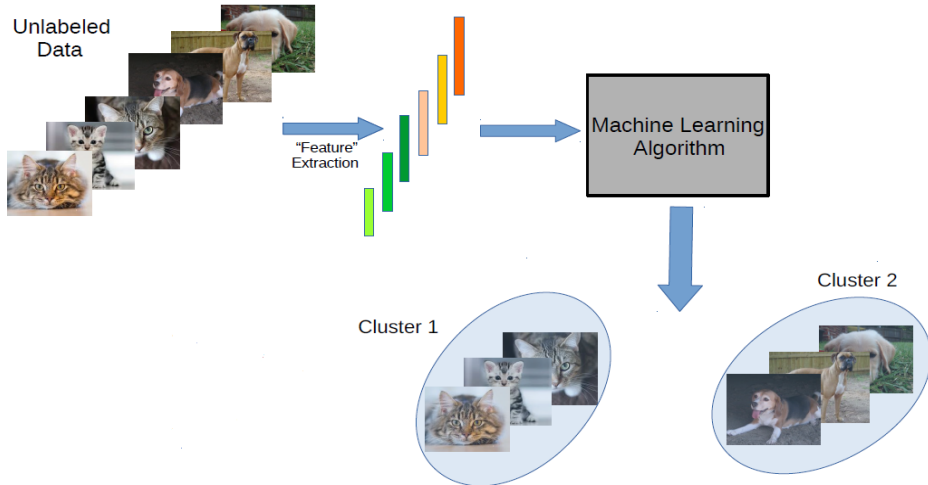
Classification: Learn a linear/nonlinear separator (the “model”) using training data consisting of input-output pairs

- Each output is discrete-valued “label” of the corresponding input
- Use it to predict the outputs for new “test” inputs



Source: CS771A (Piyush Rai)

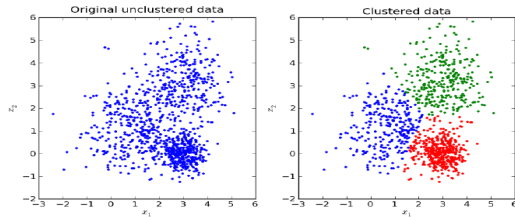
A Typical Unsupervised Learning Workflow (for clustering)



Source: CS771A (Piyush Rai)

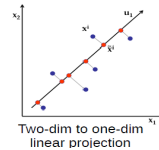
Unsupervised learning - Geometric View

Clustering: Learn the grouping structure for a given set of unlabeled inputs

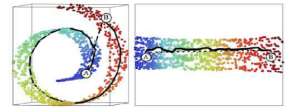


Source: CS771A (Piyush Rai)

Dimensionality Reduction: Learn a Low-dimensional representation for a given set of high-dimensional inputs

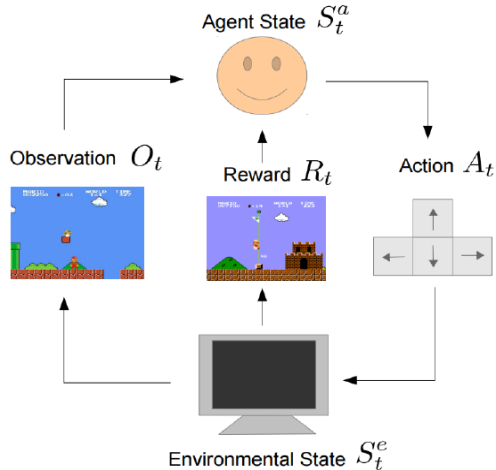


Source: CS771A (Piyush Rai)



Three-dim to two-dim
nonlinear projection

A Typical Reinforcement Learning Workflow



Agent's goal is to learn a policy for some task

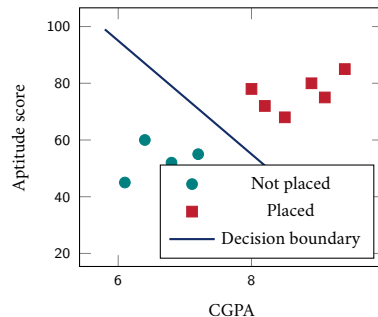
Agent does the following repeatedly

- Senses/observes the environment
- Takes an action based on its current policy
- Receives a reward for that action
- Updates its policy

Source: CS771A (Piyush Rai)

Where Does Placement Predict Fit?

- Each Student record: input features x (CGPA, internships, projects, ...)
- Outcome $y \in \{\text{Placed, Not placed}\} \Rightarrow$ **supervised**.
- Output is one of exactly **two classes** $\Rightarrow$ **binary classification**.



Why it matters

Naming the problem type correctly determines - Cost function, loss, metrics.

A Prediction Example

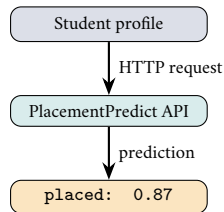
Imagine typing this into a web form and pressing **Predict**:

- CGPA: 8.4 Internships: 2
- Projects: 3 Aptitude score: 78
- Soft-skills rating: 4.2 / 5

Half a second later the server replies:

Response

```
{"placed": true, "probability": 0.87}
```



Central question of this course: what happens inside that half second — and how was that ability created?

MCQs

- In machine learning, the “rules” of the system are
 - (a) hand-written by programmers
 - (b) learned from data
 - (c) stored in a database
 - (d) fixed by the OS

MCQs

- In machine learning, the “rules” of the system are
 - (a) hand-written by programmers
 - (b) learned from data
 - (c) stored in a database
 - (d) fixed by the OS

Answer: (b)

- PlacementPredict outputs Placed / Not-placed. This is
 - (a) regression
 - (b) clustering
 - (c) binary classification
 - (d) reinforcement learning

MCQs

- In machine learning, the “rules” of the system are
 - (a) hand-written by programmers
 - (b) learned from data
 - (c) stored in a database
 - (d) fixed by the OS

Answer: (b)

- PlacementPredict outputs Placed / Not-placed. This is
 - (a) regression
 - (b) clustering
 - (c) binary classification
 - (d) reinforcement learning

Answer: (c)

- Which paradigm uses data without labels?
 - (a) supervised
 - (b) unsupervised
 - (c) reinforcement
 - (d) semi-automatic

MCQs

- In machine learning, the “rules” of the system are
(a) hand-written by programmers (b) learned from data (c) stored in a database
(d) fixed by the OS

Answer: (b)

- PlacementPredict outputs Placed / Not-placed. This is
(a) regression (b) clustering (c) binary classification (d) reinforcement learning

Answer: (c)

- Which paradigm uses data without labels?
(a) supervised (b) unsupervised (c) reinforcement (d) semi-automatic

Answer: (b)

- Predicting a student's expected salary in rupees is
(a) classification (b) clustering (c) regression (d) recommendation

MCQs

- In machine learning, the “rules” of the system are
(a) hand-written by programmers (b) learned from data (c) stored in a database
(d) fixed by the OS

Answer: (b)

- PlacementPredict outputs Placed / Not-placed. This is
(a) regression (b) clustering (c) binary classification (d) reinforcement learning

Answer: (c)

- Which paradigm uses data without labels?
(a) supervised (b) unsupervised (c) reinforcement (d) semi-automatic

Answer: (b)

- Predicting a student's expected salary in rupees is
(a) classification (b) clustering (c) regression (d) recommendation

Answer: (c)

MCQs

- The pair (x_i, y_i) in supervised learning denotes
 - (a) two features
 - (b) feature vector and label
 - (c) two labels
 - (d) model and loss

MCQs

- The pair (x_i, y_i) in supervised learning denotes
 - (a) two features (b) feature vector and label (c) two labels (d) model and loss
- “Learning” formally means
 - (a) memorising the dataset (b) minimising a loss over parameters (c) increasing dataset size (d) compiling code

Answer: (b)

MCQs

- The pair (x_i, y_i) in supervised learning denotes
(a) two features (b) feature vector and label (c) two labels (d) model and loss

Answer: (b)

- “Learning” formally means
(a) memorising the dataset (b) minimising a loss over parameters (c) increasing dataset size (d) compiling code

Answer: (b)

- Which happens *most frequently* in a deployed system?
(a) training (b) hyperparameter search (c) inference (d) data labelling

MCQs

- The pair (x_i, y_i) in supervised learning denotes
(a) two features (b) feature vector and label (c) two labels (d) model and loss

Answer: (b)

- “Learning” formally means
(a) memorising the dataset (b) minimising a loss over parameters (c) increasing dataset size (d) compiling code

Answer: (b)

- Which happens *most frequently* in a deployed system?
(a) training (b) hyperparameter search (c) inference (d) data labelling

Answer: (c)

- An agent learning chess through rewards is an example of
(a) supervised learning (b) unsupervised learning (c) reinforcement learning (d) transfer learning

MCQs

- The pair (x_i, y_i) in supervised learning denotes
(a) two features (b) feature vector and label (c) two labels (d) model and loss

Answer: (b)

- “Learning” formally means
(a) memorising the dataset (b) minimising a loss over parameters (c) increasing dataset size (d) compiling code

Answer: (b)

- Which happens *most frequently* in a deployed system?
(a) training (b) hyperparameter search (c) inference (d) data labelling

Answer: (c)

- An agent learning chess through rewards is an example of
(a) supervised learning (b) unsupervised learning (c) reinforcement learning (d) transfer learning

Answer: (c)

MCQs

- “Students like you also chose data science electives” is closest to
(a) binary classification (b) recommendation (c) anomaly detection (d) regression

MCQs

- ④ “Students like you also chose data science electives” is closest to
(a) binary classification (b) recommendation (c) anomaly detection (d) regression
Answer: (b)
- ④ The feature vector for a student in PlacementPredict could include
(a) CGPA and internships (b) the placement outcome (c) the loss value (d) the server's IP address

MCQs

- ❶ “Students like you also chose data science electives” is closest to
(a) binary classification (b) recommendation (c) anomaly detection (d) regression
Answer: (b)
- ❷ The feature vector for a student in PlacementPredict could include
(a) CGPA and internships (b) the placement outcome (c) the loss value (d) the server's IP address
Answer: (a)
- ❸ Grouping students into similar profiles *without* placement labels is
(a) classification (b) clustering (c) regression (d) calibration

MCQs

- ❸ “Students like you also chose data science electives” is closest to
(a) binary classification (b) recommendation (c) anomaly detection (d) regression
Answer: (b)
- ❹ The feature vector for a student in PlacementPredict could include
(a) CGPA and internships (b) the placement outcome (c) the loss value (d) the server’s IP address
Answer: (a)
- ❺ Grouping students into similar profiles *without* placement labels is
(a) classification (b) clustering (c) regression (d) calibration
Answer: (b)
- ❻ A model trained on biased historical data will
(a) automatically correct the bias (b) reflect the bias in predictions (c) refuse to predict (d) overfit less

MCQs

- ❸ “Students like you also chose data science electives” is closest to
(a) binary classification (b) recommendation (c) anomaly detection (d) regression
Answer: (b)
- ❹ The feature vector for a student in PlacementPredict could include
(a) CGPA and internships (b) the placement outcome (c) the loss value (d) the server’s IP address
Answer: (a)
- ❺ Grouping students into similar profiles *without* placement labels is
(a) classification (b) clustering (c) regression (d) calibration
Answer: (b)
- ❻ A model trained on biased historical data will
(a) automatically correct the bias (b) reflect the bias in predictions (c) refuse to predict (d) overfit less
Answer: (b)

The Lifecycle Stages

Overview of the eight stages

Stages

- 1 Problem framing
- 2 Data collection & labelling
- 3 Data preparation
- 4 Feature engineering
- 5 Model training
- 6 Evaluation & selection
- 7 Deployment / serving
- 8 Monitoring & retraining (loop back)

The Lifecycle

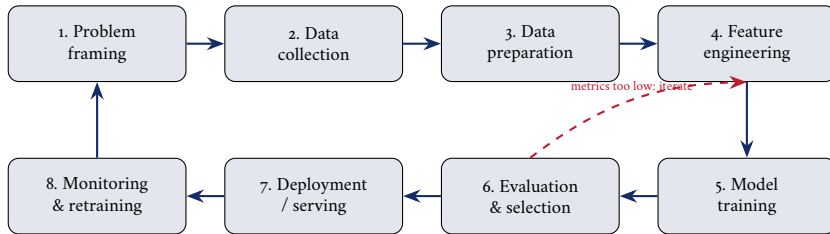


Figure: The eight-stage lifecycle. Solid arrows = normal flow; the long arrow back to Stage 1 shows that monitoring restarts the cycle; the red dashed arrow shows that poor results send you back to improve features.

Problem framing

Problem framing Turn a real-world need into a precise ML task. Decide what you are predicting, what the label is, and how you will measure success.

- Task (T): predict: Placed / Not placed for one student.
- Metric (P): F1-score (good when the two classes are unbalanced).
- Constraints: fast response, fair to all student groups.

Unbalanced classes means the two outcomes don't appear equally often in your data. Suppose out of 1000 past students, 900 got placed and only 100 did not. That's unbalanced — one class (placed) hugely outnumbers the other.

F1 score fixes this by combining two better measures: **Precision** and **Recall**. The score runs from 0 (terrible) to 1 (perfect).

Data collection & Preparation

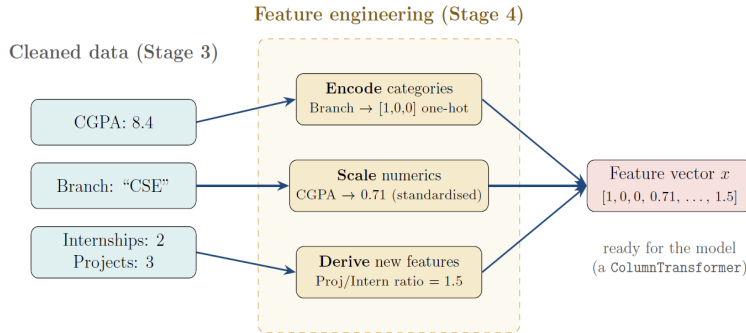
Data collection Gather raw student records and their true outcomes.

- Collect anonymised academic + co-curricular records.
- Get reliable *labels*: the actual placement result.
- Beware *sampling bias*: data from only CS students will not generalise to everyone.

Data preparation: Raw data is messy. Clean it before doing anything clever.

- Fix wrong types, remove duplicates, check outliers.
- Handle missing values (delete, or fill with mean/median).
- Split into train / validation / test before fitting.
- Guard against data leakage: no test information may sneak into training.

Feature engineering



Convert cleaned data into numbers a model can learn from.

- **Encode** categories (one-hot / ordinal encoding).
- **Scale** numeric columns so no single feature dominates.
- **Derive** new features (ratios, combinations) that reveal structure.

Model training

Let the algorithm learn the pattern from the training data.

- Start with a simple baseline (logistic regression).
- Tune hyper-parameters on the validation set (never the test set).
- Track every experiment (parameters, metrics, files)

“learning” → choosing parameters θ that minimise the average error (empirical risk) over the training data:

$$\hat{\theta} = \arg \min_{\theta} \frac{1}{n} \sum_{i=1}^n L(y_i, f_{\theta}(x_i)),$$

where L is a *loss* that punishes wrong predictions classification).

Evaluation & Deployment

Evaluation & selection Measure how good the model really is on data it has never seen.

- Use the right metric (F1, ROC-AUC for this project).
- Compare candidate models on the same fair protocol.
- Check subgroups (branches, genders) for fairness.
- Touch the *test* set only once, at the very end, for the final honest number.

Deployment: Put the chosen model to work so real users can call it.

- Package the model together with its fitted feature pipeline.
- Expose it as a REST endpoint (FastAPI /predict) or a batch job.
- Export a portable format (joblib, ONNX)

Monitoring & retraining

A deployed model slowly goes stale as the world changes.

- Track latency, errors and prediction quality.
- Detect *data drift* (inputs change) and *concept drift* (the relationship changes).
- Retrain on a schedule or when drift crosses a threshold.
- This closes the loop back to Stage 1.

Each Stage in PlacementPredict

Lifecycle stage	PlacementPredict component	Typical artifact
Data collection	anonymised academic + co-curricular records	placement_predict_50k.csv
Preparation	imputation, outlier checks	cleaned DataFrame
Feature engineering	scaling, one-hot encoding	ColumnTransformer
Training	fit classifier on historical labels	model.joblib / ONNX
Evaluation	ROC-AUC, calibration on held-out data	metrics report
Deployment	FastAPI REST endpoint	/predict service
Monitoring	drift dashboards, latency logs	PSI report, alerts
Retraining	scheduled or drift-triggered	new model version